# HeteroKV: Source-Cued Evidence Architecture and Activation Contract

## A Full-source VLM (conceptual reference)

All source content
OCR / layout / images → Full-source input payload → **VLM** → Answer

Visual evidence retained; active input may grow with source content.

## B Source-local text top-k (selection controls)

Same supplied source scope → Random / OCR lexical layout-aware ranking → Top-k text units → **VLM**

Bounded text payload; image pixels and visual relations are not forwarded.

## C HeteroKV: query-independent external memory + source-scoped bounded activation

### 01 Ingest & memory

Query-independent; CPU / disk

Pages / OCR blocks
Layout spans / image refs

**Content normalization**

OCR text + source coordinates

**Typed MemoryUnits**

source_id, unit_id, modality
text, span / bbox, image ref
provenance; unit confidence

**Source-local index / files**

unit_id -> typed record
token -> source-local unit IDs
CPU index; disk image files

### 02 Source-local selection

Supplied handle, not source discovery

**Question q + source handle s**
**Given by application**

**Source-handle filtering**
$U_s = \{u : source(u) = s\}$

**Lexical candidate union**
$C_s = \bigcup_{t \in tok(q)} inv_s(t)$
No hit: source-local text fallback

**Deterministic fusion score**
$\sigma_{base}(q,u) = w^T[L,H,M,A,C]$
w = [0.38, 0.34, 0.12, 0.08, 0.08]
Heuristic; not a learned encoder

**Document / answer rerank**

Top-3k -> document top-k -> reorder
Query / unit cues; stable ID tie break

$I_s$

### 03 Evidence-family budgets

B2 and B0 are distinct operating points

**B2 Source-image payload**

One routed source image + question
One image; processor pixel cap
Independent of text-unit ranking
$P_{B2} = Pack_{image}(I_s, q; P_{max})$

**Optional conservative hybrid**

Units / regions -> heuristic gate
Sufficient: unit-based payload
Insufficient: B2 source-image fallback

Different text/image cost; not equal payload

**B0 Text-only payload**

At most k OCR / layout units; no image
E3: 520 chars / block; 2,200 combined
Common instruction + question added
$E_k = Top_k(R(q,U_s))$
$P_{B0} = Clip_{C_text}(Serialize(E_k)) + q$

### 04 VLM request boundary

Only packed evidence is activated

**EvidencePacket assembly**

Selected unit / image references
source_id, unit_id, span / bbox
Scores + breakdown (text route)
Source / transform provenance
Payload family + active counts

**Chat / vision processor**

Text + selected image, if present

**Qwen2-VL generator**

Local / shared; no weight updates

Answer + provenance / accounting trace
No unselected record bypass to the VLM

### Score features (heuristic, not learned)

L: lexical; H: SHA-1 text-hash cosine; M: metadata
A: history if recorded; C: stored confidence
Current hash route does not encode image pixels.

### Residency contract: active input vs external storage

$M_{system} = M_{model} + M_{active}(P) + M_{external}(U) + M_{runtime}$
Input caps are fixed; external CPU / disk storage can grow.

### Observed request accounting

Active units / images / tokens; payload bytes
GPU allocated / reserved; latency
Provenance is traceability, not a security proof.

External data / storage · Control / service · B0 text payload · B2 image payload · Solid: evidence · Dashed: control · Dotted: accounting